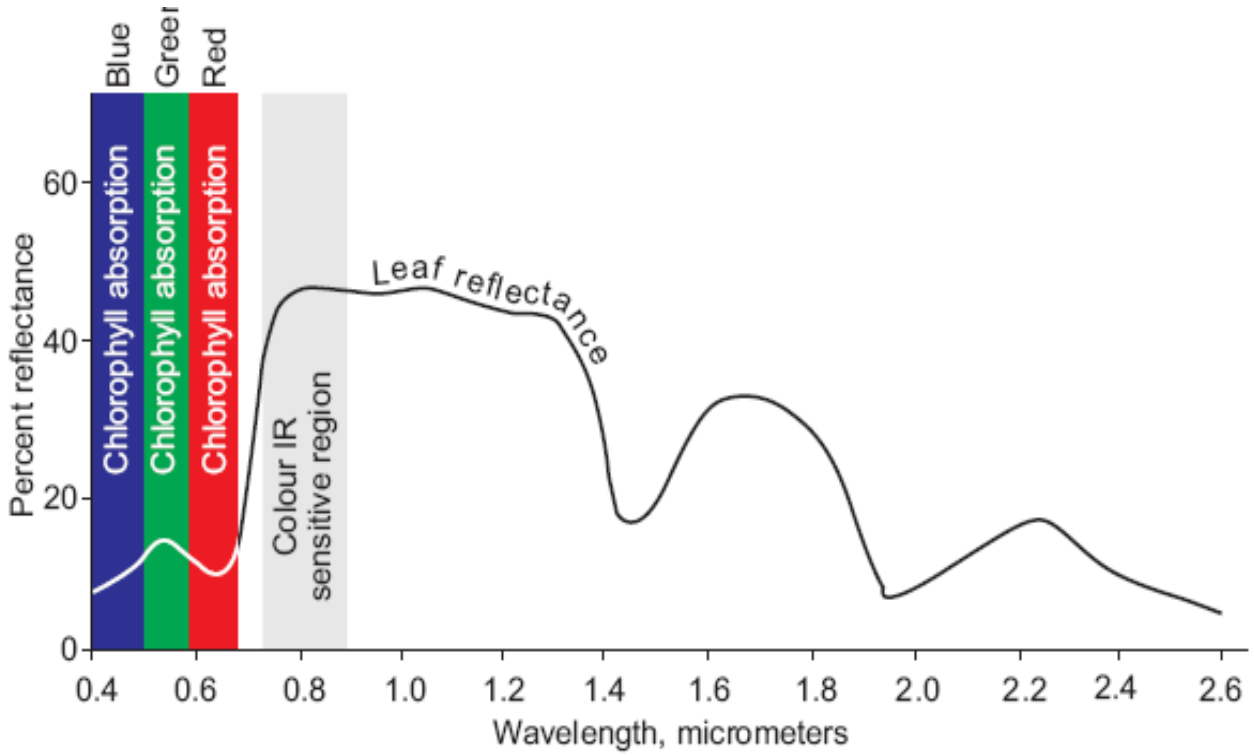
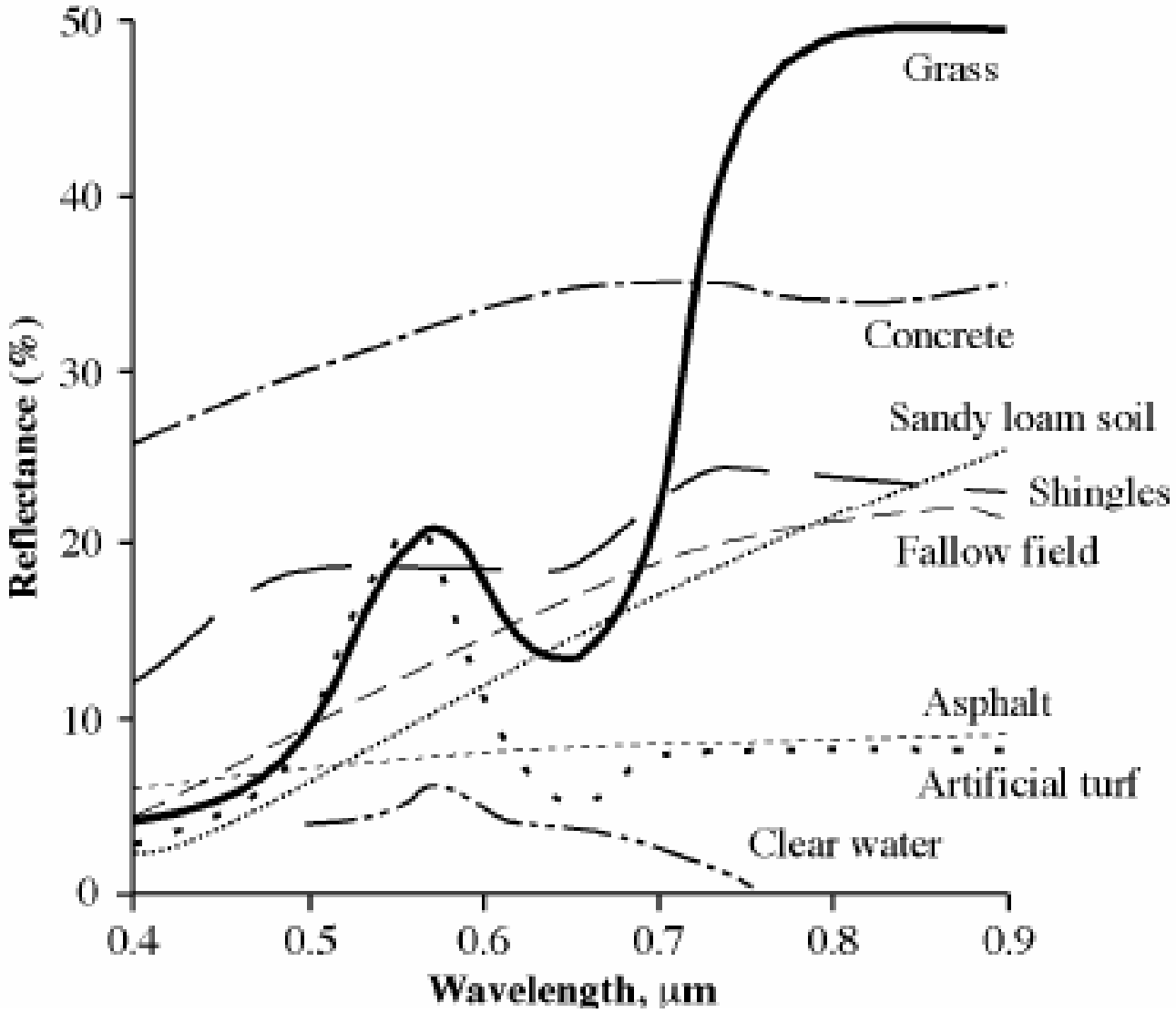

بِسْمِ اللَّهِ الرَّحْمَٰنِ الرَّحِيمِ

Spectral Reflectance of Vegetation



- Strong absorption in visible (due to Chlorophyll concentration, high content more absorption, least reflectance)
- High reflectance in NIR (live vegetation)
- At Senescence or Stress, Red reflectance increase, and NIR decreases
- Temporal Effects

Spectral Curves- Land Covers



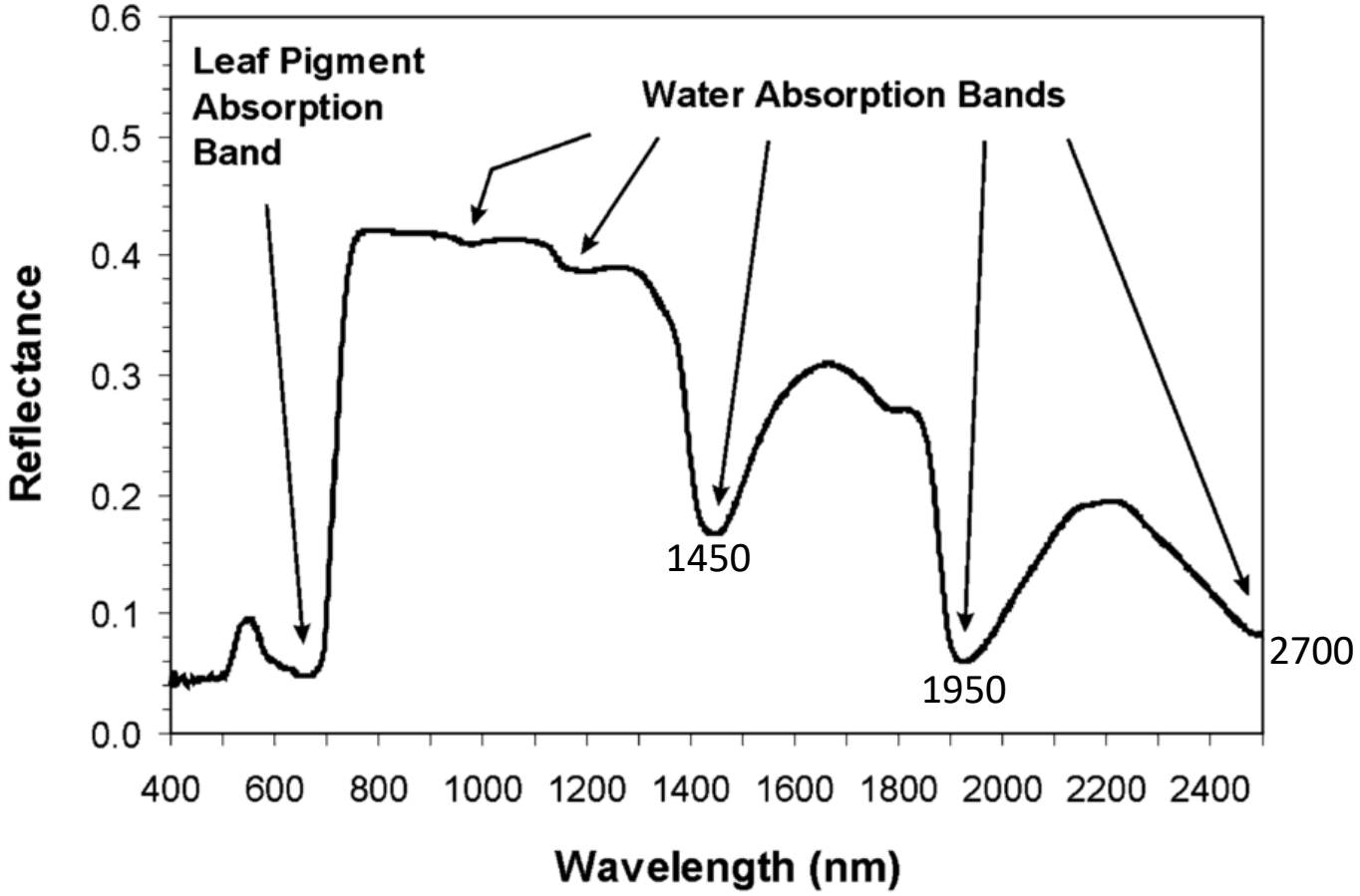
Spectral Reflectance Curves

- Curve exhibits the "peak-and-valley" configuration
- High amount of reflectance of a wavelength from a particular feature may result in peaks in the graph & low reflectance results in a dip or valley in the curve
- In other words, the peaks indicate strong reflection of incident energy and
- The valleys indicate predominant absorption of the energy in the corresponding wavelength bands

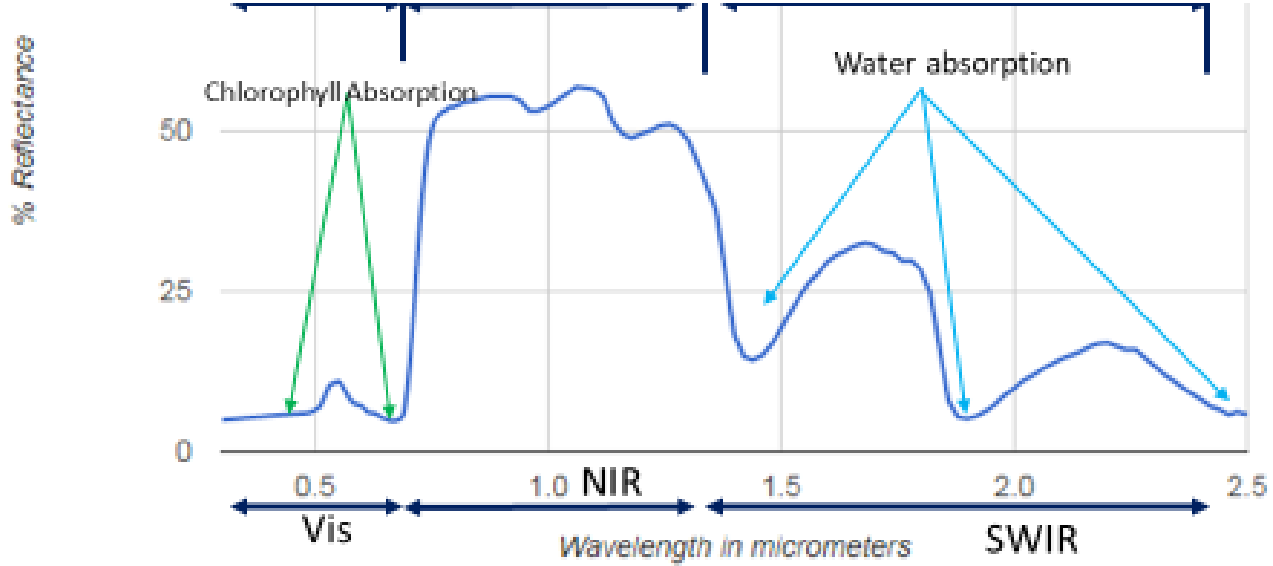
Spectral Reflectance Curves

- The peaks & dips in these curves are very specific, at definite wavelengths, so should be carefully & correctly drawn at exact wavelength (μm)
- The energy/matter interactions at visible wavelengths are very complex and depend on several interrelated factors (water)

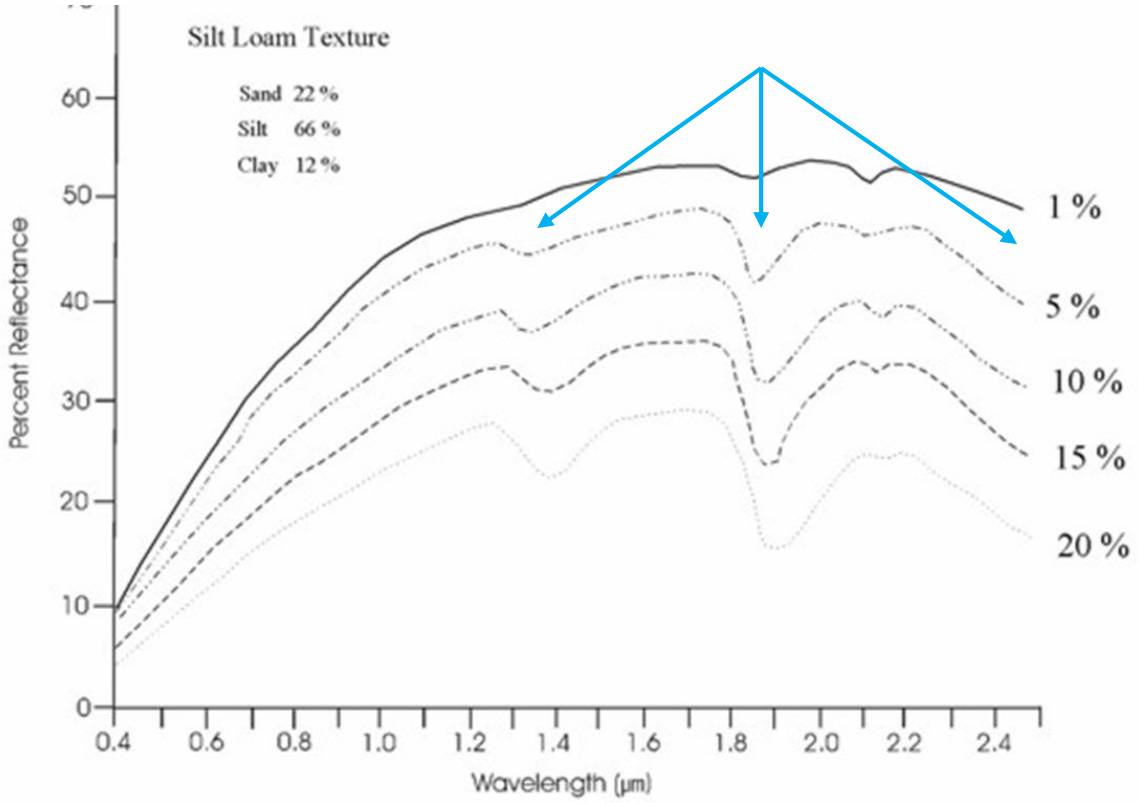
Spectral Absorption



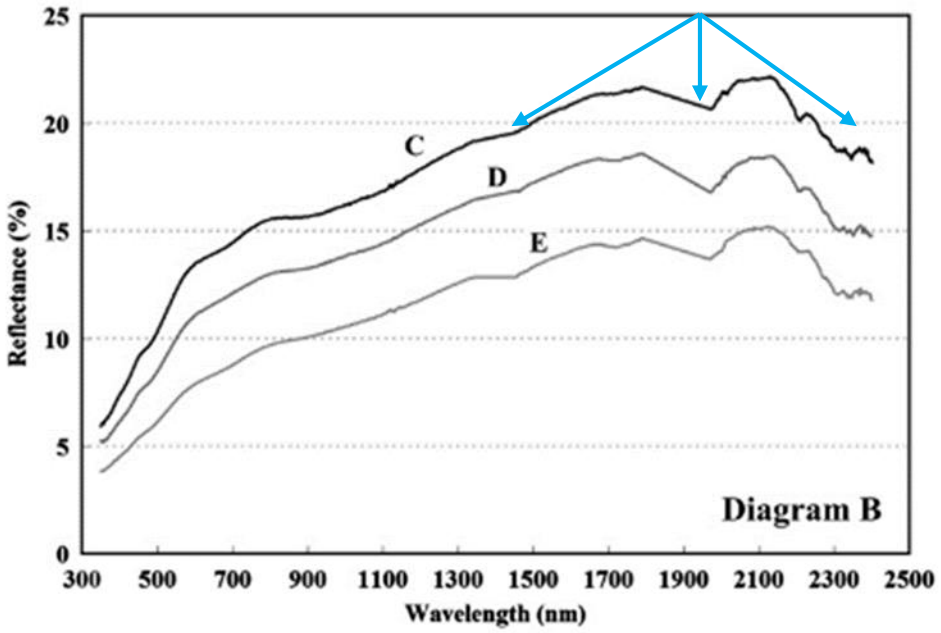
Water absorption bands: The part of the EM spectrum (specific infrared wavelengths) that are strongly absorbed by water vapours

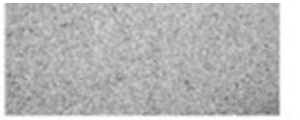


Vegetation

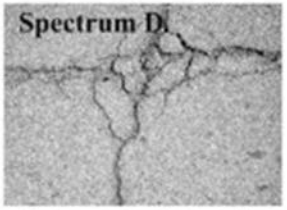


Soil

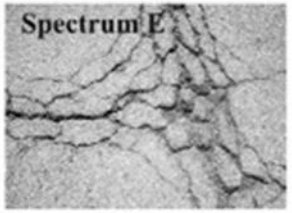




Age:>10yrs/PCI:32/SI:63

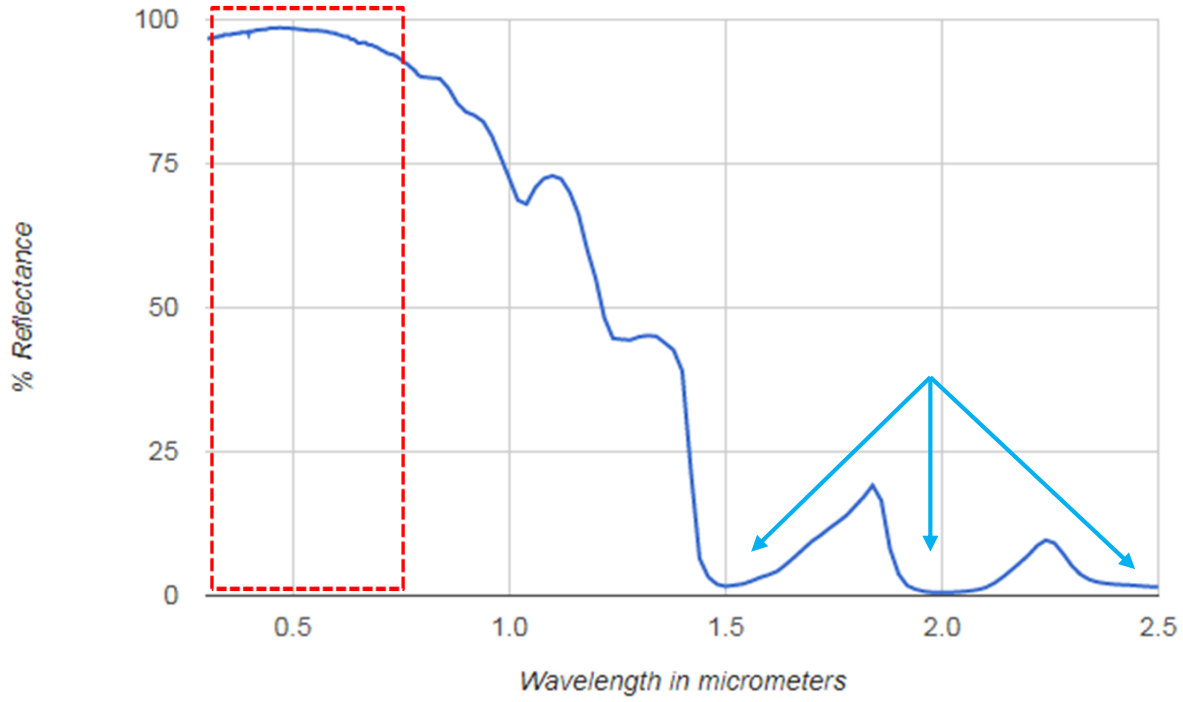


Low severity crack



Severe alligator crack

Roads



Snow and Ice

Reference and Reading Assignment

- Paul M Mather, Chapter 1, section 1.3.2 Spectral reflectance of Earth surface materials
- Journal papers -LMS

Reference and Reading Assignment

- *USGS Spectral Library Johns Hopkins (JHU)*

  Spectral Library

- *DLR and GFZ Laboratory Spectra*

Reference and Reading Assignment

- Paul M Mather, Chapter 1, section 1.3.2 Spectral reflectance of Earth surface materials
- Journal papers -LMS

